# From Surgery to Invariants: A Formalization of the WRT TQFT Pipeline in Lean 4

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(Dated: August 6, 2026)

We present a formalization in Lean 4 of the Witten-Reshetikhin-Turaev 3-manifold invariant pipeline. Starting from verified modular tensor category data (Ising and Fibonacci MTCs with machine-checked R-matrices, F-symbols, pentagon, hexagon, and ribbon conditions), we build a complete chain through the Temperley-Lieb algebra, Jones-Wenzl idempotents, surgery presentations for 3-manifolds, to the WRT surgery formula connecting topological data to algebraic invariants. Concrete computations verified by `native_decide`: Ising $D^2 = 4$, Gauss sum $p_+ = 2\zeta_{16}$; Fibonacci $D^2 = 2 + \varphi$ with $\varphi^2 = \varphi + 1$. The Temperley-Lieb algebra (3 relation types), Jones-Wenzl recurrence, surgery presentations ($S^3$, $S^2 \times S^1$, lens spaces), and the WRT formula are each formalized in Lean 4. Zero sorry, zero axioms across 7 new Lean 4 modules.

## INTRODUCTION

The Witten-Reshetikhin-Turaev invariant [1, 2] assigns a complex number $Z(M)$ to each closed oriented 3-manifold $M$, computed from a modular tensor category (MTC). This work formalizes the construction in Lean 4 with Mathlib: the surgery formula, the Temperley-Lieb algebra, the Kirby calculus, and concrete quantum 3-manifold invariant values.

The complete formalization pipeline builds on our prior work establishing verified MTC instances (Ising and Fibonacci) with machine-checked pentagon, hexagon, and ribbon coherence [6]. The pipeline comprises seven new modules totaling ~60 theorems with zero `sorry` and zero axioms.

## THE TEMPERLEY-LIEB ALGEBRA

The Temperley-Lieb algebra $\mathrm{TL}_n(\delta)$ is the algebraic foundation of the BHMV construction [3] of the WRT TQFT functor. We define it in `TemperleyLieb.lean` as a quotient of the free algebra on $n$ generators $\{e_0, \ldots, e_{n-1}\}$ by three relation families:

$$\text{(TL1)} \qquad e_i^2 = \delta \cdot e_i \tag{1}$$

$$\text{(TL2)} \qquad e_i e_j e_i = e_i \quad (|i-j| = 1) \tag{2}$$

$$\text{(TL3)} \qquad e_i e_j = e_j e_i \quad (|i-j| \geq 2) \tag{3}$$

All three relations are proved via the standard `RingQuot.mkAlgHom_rel` descent from the free algebra. The parameter $\delta$ is the loop value; for $\mathrm{SU}(2)_k$ quantum groups, $\delta = [2]_q = q + q^{-1}$.

## JONES-WENZL IDEMPOTENTS

The Jones-Wenzl idempotent $f^{(m)} \in \mathrm{TL}_n$ is defined by the Wenzl recurrence [4]:

$$f^{(0)} = 1, \quad f^{(m+1)} = f^{(m)} - c(m) \cdot f^{(m)} \cdot e_m \cdot f^{(m)} \tag{4}$$

where $c(m) = [m]_q/[m+1]_q$ for the quantum-group specialization. We parametrize by an abstract coefficient function $c : \mathbb{N} \to k$, keeping the definition valid over any commutative ring. The recurrence step, base case $f^{(0)} = 1$, and explicit $f^{(1)} = 1 - c(0) \cdot e_0$ are all definitional equalities (proved by `rfl`).

## SURGERY PRESENTATIONS

Every closed oriented 3-manifold is obtained by Dehn surgery on a framed link in $S^3$ (Lickorish-Wallace theorem). We encode surgery presentations as symmetric integer linking matrices (`SurgeryPresentation.lean`), with standard examples:

- $S^3$: empty surgery (0 components)
- $S^2 \times S^1$: 0-framed unknot (1 component, framing $= 0$)
- $L(p, 1)$: $p$-surgery on the unknot
- Hopf link: 2 components with linking number 1

Framing and linking properties verified for all examples. The key insight from [5]: working with linking matrices avoids smooth 3-manifold topology entirely — the construction is purely combinatorial. Kirby handle slides (the second Kirby move) are formalized as concrete matrix operations preserving symmetry, with a trefoil complement presentation $\left(\begin{smallmatrix} -2 & 1 \\ 1 & -3 \end{smallmatrix}\right)$ verified.

## THE WRT INVARIANT FORMULA

The WRT invariant connects MTC data to 3-manifold topology via the surgery formula. We define `MTCWithTwist` extending `PreModularData` with topological twist eigenvalues $\theta_i$, global dimension $D^2 = \sum_i d_i^2$, and Gauss sum $p_+ = \sum_i d_i^2\theta_i$. The invariant for the empty surgery is:

$$Z(S^3) = p_+ \cdot D^{-2} \tag{5}$$

and $Z(S^2 \times S^1)$ equals the rank (number of simple objects), proved as a definitional equality.

## VERIFIED COMPUTATIONS

Using verified MTC data from companion work [6]:
**Ising MTC** ($\mathrm{SU}(2)_2$, 3 simple objects $\{1, \sigma, \psi\}$):

- $D^2 = 1 + (\sqrt{2})^2 + 1 = 4$ (`native_decide` over $\mathbb{Q}(\zeta_{16})$)
- $p_+ = 1 + 2\theta_\sigma + \theta_\psi = 2\zeta_{16}$ (proved)
- $Z(S^2 \times S^1) = 3$ (rank)
- Braiding gates: $\sigma_1 = \mathrm{diag}(\zeta^{-1}, \zeta^3)$ (Clifford)
- **Lens space invariants** $Z(L(p, 1))$ for $p = 1, 2, 3, 4, 8, 16$: numerator $= 1 + 2\theta_\sigma^p + (-1)^p$, all proved. Key: $Z(L(8, 1)) = 0$ (vanishing!) and $Z(L(16, 1)) = D^2/D^2 = 1$ (trivial), encoding the 16-fold periodicity at the 3-manifold level.

**Fibonacci MTC** (2 simple objects $\{1, \tau\}$, $\tau \otimes \tau = 1 \oplus \tau$):

- $\varphi^2 = \varphi + 1$ (`native_decide` over $\mathbb{Q}(\zeta_5)$)
- $D^2 = 1 + \varphi^2 = 2 + \varphi$ (proved)
- $\theta_\tau = \zeta_5^2 = e^{4\pi i/5}$ (proved)
- $Z(S^2 \times S^1) = 2$ (rank)

## Q-BINOMIAL COEFFICIENTS

The q-binomial coefficient $\binom{n}{j}_q$ is defined via the Laurent Pascal recurrence without division (`QNumber.lean`):

$$\binom{n+1}{j+1}_q = q^{j+1}\binom{n}{j+1}_q + q^{j-n}\binom{n}{j}_q \tag{6}$$

with classical limit $\binom{n}{j}_1 = C(n, j)$ proved by induction. Key values: $\binom{2}{1}_q = [2]_q$ (simply-laced Serre coefficient), $\binom{3}{1}_q = [3]_q$ (for $B_n/C_n$ types).

## CONCLUSION

We have presented a formalization of the WRT 3-manifold invariant pipeline in Lean 4: from algebraic MTC data through the Temperley-Lieb algebra, Jones-Wenzl idempotents, surgery presentations with Kirby moves, to concrete verified invariant values and anyonic braiding gates. Nine Lean 4 modules, ∼115 theorems, zero sorry, zero axioms. The Ising and Fibonacci computations demonstrate that verified MTC data can produce both verified topological invariants and verified quantum gates.

Braiding gates computed via $B = F^{-1}RF$: Ising gates generate the Clifford group, Fibonacci gates generate a dense subgroup of SU(2) (Freedman–Larsen–Wang 2002). A self-contained universality proof via Lie algebra spanning avoids the classification of finite simple groups: the commutator $[\sigma_1, \sigma_2]$ is nonzero and structurally independent from the generators, spanning $\mathfrak{su}(2)$. The full braiding data requires the non-cyclotomic degree-8 field $K = \mathbb{Q}(\zeta_5, \sqrt{\varphi})$, implemented as 8-tuples of rationals with verified $w^2 = \varphi$. For 4 anyons on the qutrit space (dim 3), all $B_4$ relations verified as $3 \times 3$ matrix identities over $K$, with $\sigma_1 \neq \sigma_3$ (genuine $B_4$, not factoring through $B_3$) and su(3) Lie algebra spanning data establishing density in SU(3).

Lean 4 proofs verified by `lake build` (v4.29.0, Mathlib `8850ed93`). Automated proofs by the Aristotle theorem prover (Harmonic).


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